

# FTML practical session 6

3 juin 2026

## 1 LINE SEARCH FOR LEAST SQUARES

We first note that

$$\begin{aligned}\nabla_{\theta} f(\alpha(\gamma)) &= H\alpha(\gamma) - \frac{1}{n}X^T y \\ &= H(\theta_t - \gamma \nabla_{\theta} f(\theta_t)) - \frac{1}{n}X^T y \\ &= \nabla_{\theta} f(\theta_t) - \gamma H \nabla_{\theta} f(\theta_t)\end{aligned}\tag{1}$$

Let us derivate  $g(\gamma) = f(\alpha(\gamma))$  with respect to  $\gamma$ . By a composition :

$$\begin{aligned}g'(\gamma) &= \langle \nabla_{\theta} f(\alpha(\gamma)), \alpha'(\gamma) \rangle \\ &= -\langle \nabla_{\theta} f(\theta_t) - \gamma H \nabla_{\theta} f(\theta_t), \nabla_{\theta} f(\theta_t) \rangle \\ &= -\|\nabla_{\theta} f(\theta_t)\|^2 + \gamma \langle H \nabla_{\theta} f(\theta_t), \nabla_{\theta} f(\theta_t) \rangle\end{aligned}\tag{2}$$

In order to cancel the derivative, we must have that

$$\gamma^* = \frac{\|\nabla_{\theta_t} f\|^2}{\langle H \nabla_{\theta} f(\theta_t), \nabla_{\theta} f(\theta_t) \rangle}\tag{3}$$

We note that this is correct if  $\nabla_{\theta} f(\theta_t) \neq 0$ . If  $\nabla_{\theta} f(\theta_t) = 0$ , this means that  $\theta_t = \eta^*$ , as  $f$  is convex.

This computation may then be done at each iteration.

An important remark is that if we note  $\theta_{t+1}^* = \theta_t - \gamma^* \nabla_{\theta} f(\theta_t) = \alpha(\gamma^*)$ , then equations 1 and 2 shows that

$$\langle \nabla_{\theta} f(\theta_{t+1}^*), \nabla_{\theta} f(\theta_t) \rangle = 0\tag{4}$$

Two optimal directions of the gradient updates are **orthogonal**. Importantly, this is true in the general case, not only for least-squares.

### 1.0.1 Backtracking line search

In many practical situations, it is not possible to compute explicitly the optimal step  $\gamma^*$ . Or it could be possible, but too expensive computationnally.

In such situations, it is possible to compute an approximation of  $\gamma^*$ , for instance using **backtracking line search**. This method attempts to find a good  $\gamma$  by trying several decreasing values until a sufficient decrease in  $f$  after the gradient update is obtained.

[https://en.wikipedia.org/wiki/Backtracking\\_line\\_search](https://en.wikipedia.org/wiki/Backtracking_line_search)